



Group 6

Metagenomic Analysis of Microbiomes

**How does the microbiome differ between patients
with type 2 diabetes and healthy controls?**

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Introduction

Diabetes is a chronic disease that affects millions of people worldwide, where approximately 95% of diabetes cases are classified as type 2 diabetes (T2D) [1]. The disease is characterized by insulin resistance and progressive beta-cell dysfunction, resulting in hyperglycemia and T2D [2]. This project investigates whether the composition of the gut microbiome differs between individuals with T2D and healthy control individuals. To minimize the confounding effects of geographical variation, part of the analysis was restricted to a Chinese cohort. Because microbiome data are sparse, zero-inflated (Figure S1 a, b, and c), and compositional, multiple statistical approaches were applied, including non-parametric tests, linear models, with and without CLR transformations, as well as multivariate analyses. Combining these methods provides greater confidence in finding the true differences in gut microbial composition between individuals with T2D and healthy controls. In the downstream analysis, a p-value (or q-value) below 0.05 is considered statistically significant. When multiple tests were performed simultaneously, the Benjamini–Hochberg procedure was used to adjust p-values.

Analysis

This analysis is based on the provided phyloseq objects. Initially, the genus phyloseq object was used to get a broad overview of the data. Subsequent analysis was carried out using both the species and genus phyloseq objects.

Analysis of the full dataset on genus level

Preliminary exploration of the metadata revealed that there were 432 samples, of which 260 samples were from individuals with T2D (MESH ID: D003924) and 172 samples were from healthy controls (MESH ID: D006262)(S9 Table 1). The alpha diversity between the two groups on the genus level was investigated using the Shannon diversity index (Figure 1a). This test showed that patients had a statistically significantly higher alpha diversity with a p-value of $3.58 \cdot 10^{-10}$. Differences in beta diversity between T2D and controls were assessed using the Bray-Curtis dissimilarity metric, which was also statistically significant ($p \leq 0.001$) (Figure 2a). However, the beta-dispersion test was also highly statistically significant ($p = 4.021 \cdot 10^{-06}$) (Figure S2), indicating that part of the variance detected by the PERMANOVA could be a result of differences in dispersion rather than a true centroid shift.

The metadata furthermore showed that the 432 samples were pooled from three different projects: one from Europe (PRJEB1786), one from the United States of America (PRJNA299502) and one from China (PRJNA422434) (S10 Table 2). Interestingly, the dataset was not balanced since the Chinese cohort contained both individuals with T2D and healthy controls, whereas the two other cohorts only contained individuals with T2D (S11 Table 3). This observation was investigated further because it potentially could represent a confounding factor affecting the microbial comparisons between individuals with T2D and healthy controls. The Shannon diversity analysis on genus level showed that both the European and the American cohorts had a statistically significant higher alpha diversity than the Chinese cohort ($p < 2 \cdot 10^{-16}$ and $p = 0.00012$, respectively) (Figure 1c). Since the European and American cohorts consisted exclusively of individuals with T2D, this may have affected the observed difference in alpha diversity between the two health conditions (Figure 1a). The Bray-Curtis dissimilarity analysis furthermore showed a statistically significant difference in beta diversity between the different cohorts ($p \leq 0.001$). Although the beta-dispersion test remained significant ($p = 0.02396$) (Figure S3), it was less significant than that observed for the comparison between the two health states.

Overall, these analyses suggest that differences in cohort may act as a confounding factor, potentially influencing conclusions regarding whether and how the gut microbial community differs between T2D patients and healthy controls. Considering the technical variance that might have been introduced by the fact that the data was collected, sequenced and processed differently across the three project IDs, as well as genetics, diet and lifestyle differing across continents, and importantly because controls are only present in the Chinese cohort, downstream analysis was restricted to only the Chinese cohort (PRJNA422434).

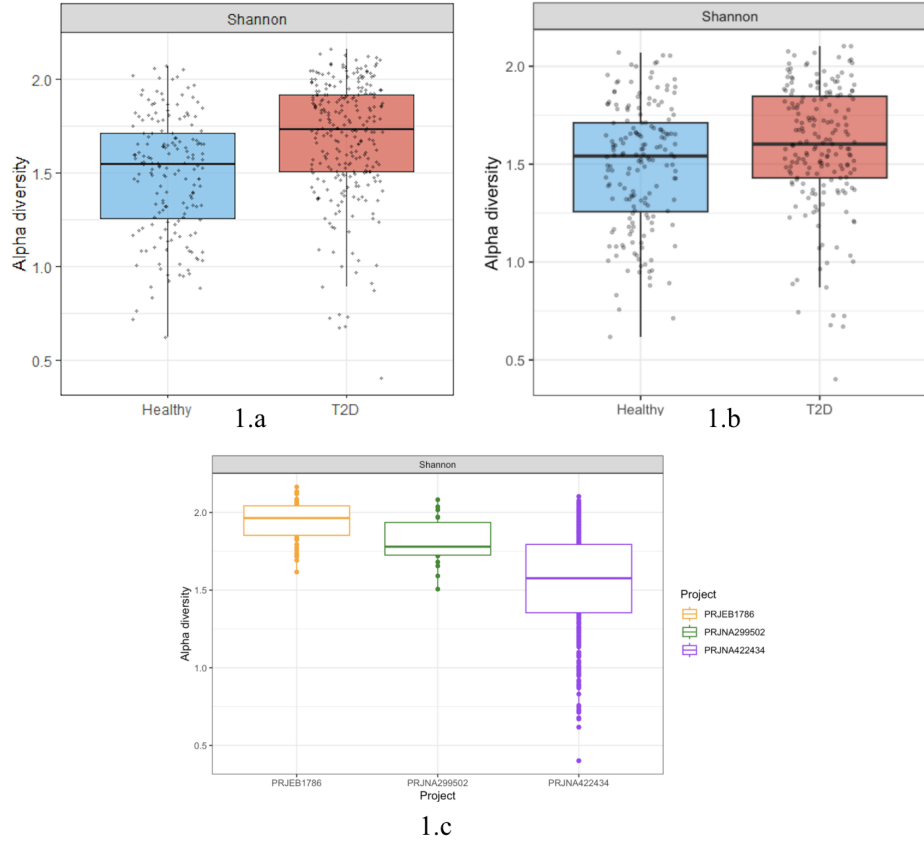


Figure 1: Alpha diversity.

The Shannon diversity index was used to investigate alpha diversity. a) Boxplot showing the difference in alpha diversity between individuals with T2D and controls. The analysis was performed on the unfiltered genus level. b) Boxplot showing the difference in alpha diversity between individuals with T2D and controls. The analysis was performed on the Chinese cohort genus level. c) Boxplot showing the difference in alpha diversity between project IDs. The analysis was performed on the unfiltered genus level. Wilcoxon and Kruskal-Wallis tests were used to investigate differences in alpha diversity.

Analysis of the Chinese cohort on genus level

The subset with only the Chinese cohort consisted of 359 samples, of which 187 were from individuals with T2D and 172 were from healthy controls. The metadata was further explored to assess whether the two disease groups were balanced with respect to their covariates. No samples had recent antibiotic use, which is favourable as antibiotics are among the strongest disruptors of gut microbial composition [3]. The disease groups did not differ significantly with respect to BMI ($p = 0.34$, metadata available for 58% of samples) and sex ($p = 0.05071$), however, T2D individuals were significantly older than controls (median 55 vs 44 years, $p = 3.511 \cdot 10^{-12}$).

Analysis of alpha diversity with the Shannon metric showed that the diversity remained significantly higher in T2D than in healthy controls ($p = 0.0018$), though less significant than in the full dataset (Figure 1b), which indicates that not having healthy controls from the other cohorts may have inflated the alpha diversity in T2D samples. Sequencing depth and Shannon alpha diversity were also tested for correlation as diversity estimates can be inflated by deeper sequencing. Despite sequencing depth being significantly higher in the T2D group ($p = 5.582 \cdot 10^{-05}$), no significant correlation was found between alpha diversity and sequencing depth (Spearman's $\rho = 0.06$ and $p = 0.27$) (Figure S4).

The beta diversity between the disease groups was measured by the Bray-Curtis dissimilarity metric and visualised by PCoA (Figure 2c). The PERMANOVA test showed a statistically significant difference in microbial composition between T2D and controls ($p \leq 0.001$). It also found that disease status explained 1.8% of the total variation compared to it explaining approximately 3% of the variation in the full dataset. In contrast to the full dataset, the betadispersion test was not significant ($p = 0.24$). This indicates that the observed difference in the Chinese cohort reflected a true centroid shift rather than difference in dispersion (Figure S5).

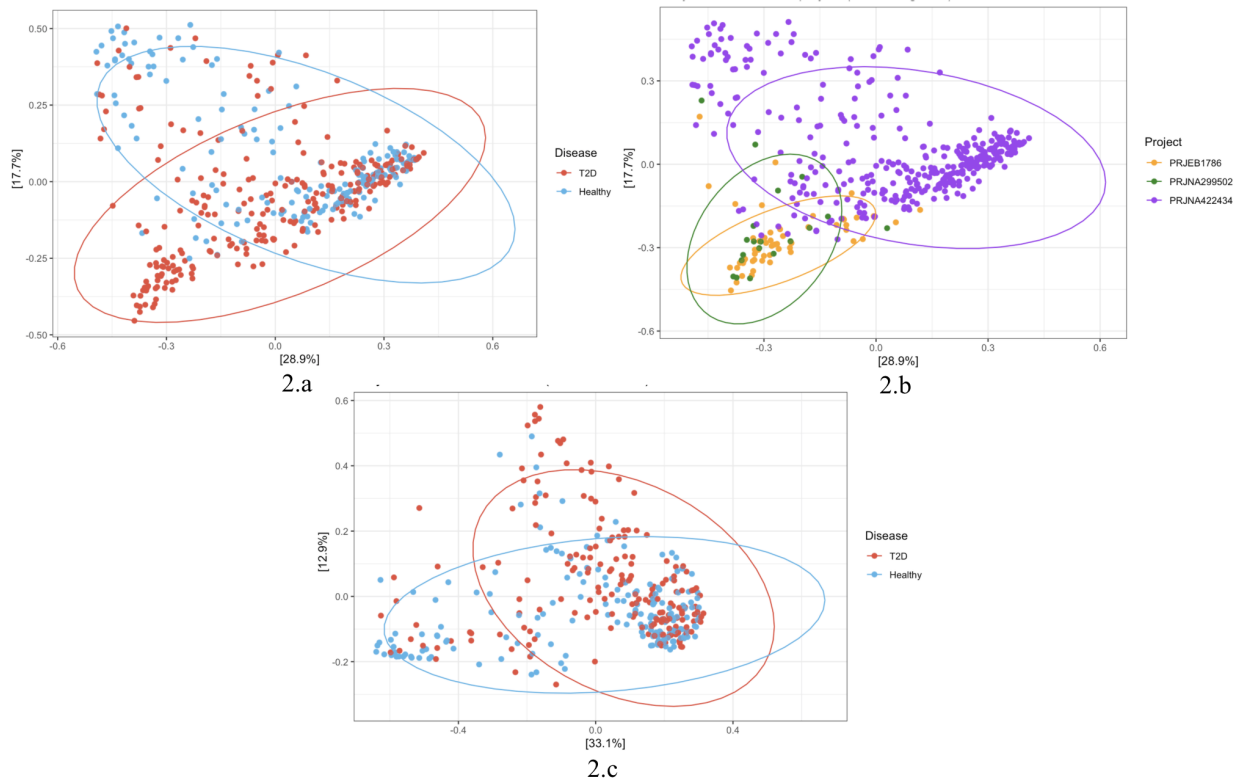


Figure 2: Beta diversity.

Beta diversity was investigated with the Bray-Curtis dissimilarity analysis. a) PCoA plot on Bray-Curtis analysis on individuals with T2D and controls. The analysis was performed on the unfiltered genus level. b) PCoA plot on Bray-Curtis analysis on different project IDs. The analysis was performed on the unfiltered genus level. c) PCoA plot on Bray-Curtis analysis from individuals with T2D and controls. The analysis was performed on the Chinese cohort genus level. A PERMANOVA (adonis2) test was used to investigate statistically significant differences followed by a beta-dispersion and ANOVA test investigating differences in dispersion.

Differentially abundant taxa on species level

The species phyloseq object was chosen for downstream analysis of differentially abundant taxa because of its improved resolution. To identify microbial taxa that differed significantly in relative abundance between individuals with T2D and healthy controls, a differential abundance analysis was performed using three different approaches. Initially, differences in relative abundance for each taxon between groups were examined using the Wilcoxon rank-sum test. Additionally, generalized linear models (GLMs) were fitted for each taxon to further evaluate the differences in abundant taxa across the two groups. Four models were considered: an unadjusted model including disease status only, a model adjusted for sex, a model adjusted for age, and a model adjusted for both sex and age. In this way it could be seen if the differential abundant taxa were affected by potential confounding variables. Furthermore, the taxa abundances were transformed using the centered log-ratio (CLR) transformation. Subsequently, the same set of linear models were used to test the differential abundances: unadjusted, adjusted for sex, adjusted for age, and adjusted for sex and age. This was done to see whether disease-associated taxa remained significant after accounting for the compositional nature of microbiome data (S6-S8).

The different statistical approaches resulted in varying significantly differentially abundant taxa. As can be seen in figure 3, several taxa were consistently enriched or depleted in individuals with T2D across multiple statistical approaches. *Roseburia intestinalis*, *Prevotella stercorea* and *Faecalibacterium duncaniae* were identified as significantly depleted in individuals with T2D by multiple statistical approaches (Figure 3). However, when adjusting for age, only *Roseburia intestinalis* remained significantly depleted in individuals with T2D. This suggests that age may partly explain the observed associations between *Prevotella stercorea* and *Faecalibacterium duncaniae* and disease status. Among taxa enriched in individuals with T2D, *Enterocloster bolteae* was significant across 5 out of 9 approaches, however it was not statistically significant when adjusting for age.

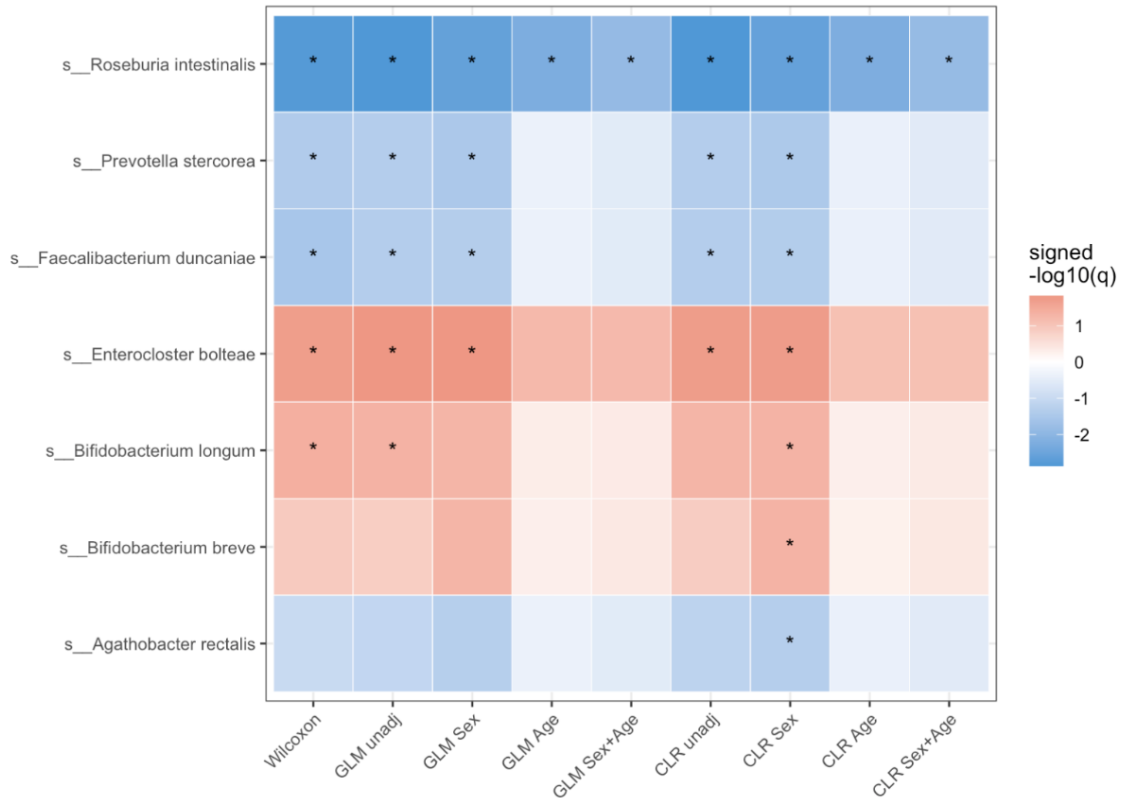


Figure 3: Heatmap showing statistically significant taxa across different statistical tests.

Heatmap summarising the results from different statistical tests investigating differentially abundant taxa between individuals with T2D and controls. The following statistical tests were performed: Wilcoxon rank-sum test, generalized linear models (unadjusted, adjusted for sex, adjusted for age and adjusted for sex and age) and linear models on CLR transformed data (unadjusted, adjusted for sex, adjusted for age, adjusted for sex and age). The asterisk indicates statistical significance. Red: enriched in individuals with T2D. Blue: depleted in individuals with T2D.

Redundancy analysis of microbial composition on species level

In order to analyse the microbiomes as whole communities rather than individual taxa, a redundancy analysis (RDA) was made (Figure 4). This was also done to look at the extent to which the variation in the microbial composition could be explained by disease status, age and sex, respectively. An ANOVA test showed that disease status was significantly associated with overall microbial community composition ($p \leq 0.001$). Sex also explained a significant proportion of the variation ($p = 0.036$), however age was not significantly associated ($p = 0.168$). Although the overall explained variation of the model was only 1.3%, a partial separation between healthy individuals and individuals with T2D was observed along the first RDA axis. Disease status had the most significant association with the microbial community composition. Disease status and age vectors were pointing in similar directions, suggesting that the effect by age and disease status on the microbial composition are associated (Figure 4a).

To explore which taxa were associated with the separation, the species with the largest RDA1 loadings were examined (Figure 4b). All seven taxa identified as differentially abundant by at least one test shown in figure 3 reappeared among the taxa associated with the separation on RDA1. Each taxa even appeared on the expected side, those depleted in T2D associated with health and those enriched in T2D associated with T2D in the RDA plot.

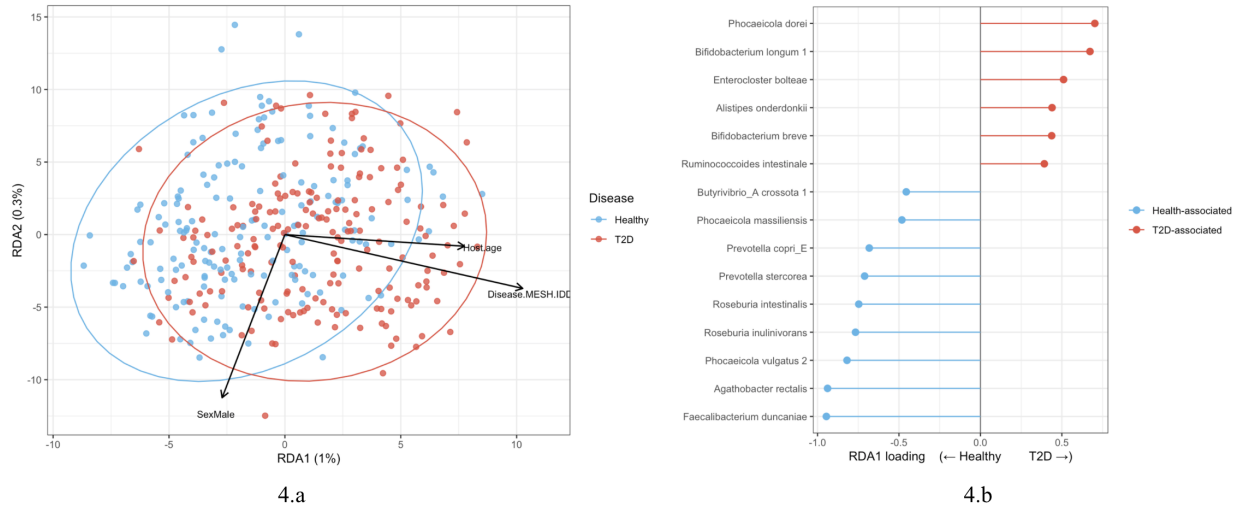


Figure 4: Multivariate statistics of the gut microbial community.

RDA analysis on CLR transformed data showing how much of the variation in the microbial community can be explained by host age, sex and disease status. An ANOVA test was used to test statistical significance. a) RDA plot constrained by disease, sex and age. Points are samples colored by disease status with 95% confidence ellipses per group. The arrows show the constraining variables. b) The 15 taxa with the largest loadings on RDA1 coloured by group.

Conclusion

This project investigated how the gut microbiome differs between individuals with T2D and healthy controls. Initial analyses revealed that the different cohorts explained more variation in the microbial composition than disease status itself, leading to further analysis being restricted to only the Chinese cohort. The study identified 7 microbial taxa that were significantly enriched or depleted in individuals with T2D across multiple statistical approaches. Overall, *Roseburia intestinalis* was the only taxon which remained statistically significant across all methods. In contrast, several other taxa changed their significance status after adjustment for age and/or sex, highlighting the importance of accounting for potential confounding variables in differential abundance analyses. Redundancy analysis demonstrated significant differences in overall community composition between disease groups. Disease status had the strongest association, sex a weaker but still significant one, whereas age was not significantly associated with community composition. This showed that age was associated with specific taxa, however it did not affect the whole microbial composition. Of the 7 taxa identified as differentially abundant via initial statistical methods, all 7 reappeared as some of the taxa associated with the separation on RDA1, underlining the importance of further investigating these taxa as potential microbial markers of T2D.

AI declaration

In this project, Generative AI was used to modify and debug R-scripts, latex formatting, and grammar questions.

References

- [1] World Health Organization, Diabetes, 2024, visited 04/06/2026
<https://www.who.int/news-room/fact-sheets/detail/diabetes>
- [2] O’Rahilly SP, Nugent Z, Rudenski AS, Hosker JP, Burnett MA, Darling P, Turner RC. Beta-cell dysfunction, rather than insulin insensitivity, is the primary defect in familial type 2 diabetes. *Lancet*. 1986 Aug 16;2(8503):360-4. doi: 10.1016/s0140-6736(86)90052-8. PMID: 2874367. <https://pubmed.ncbi.nlm.nih.gov/2874367/>
- [3] Patangia DV, Anthony Ryan C, Dempsey E, Paul Ross R, Stanton C. Impact of antibiotics on the human microbiome and consequences for host health. *Microbiologyopen*. 2022 Feb;11(1):e1260. doi: 10.1002/mbo3.1260. PMID: 35212478; PMCID: PMC8756738.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8756738>

Supplementary

S1:

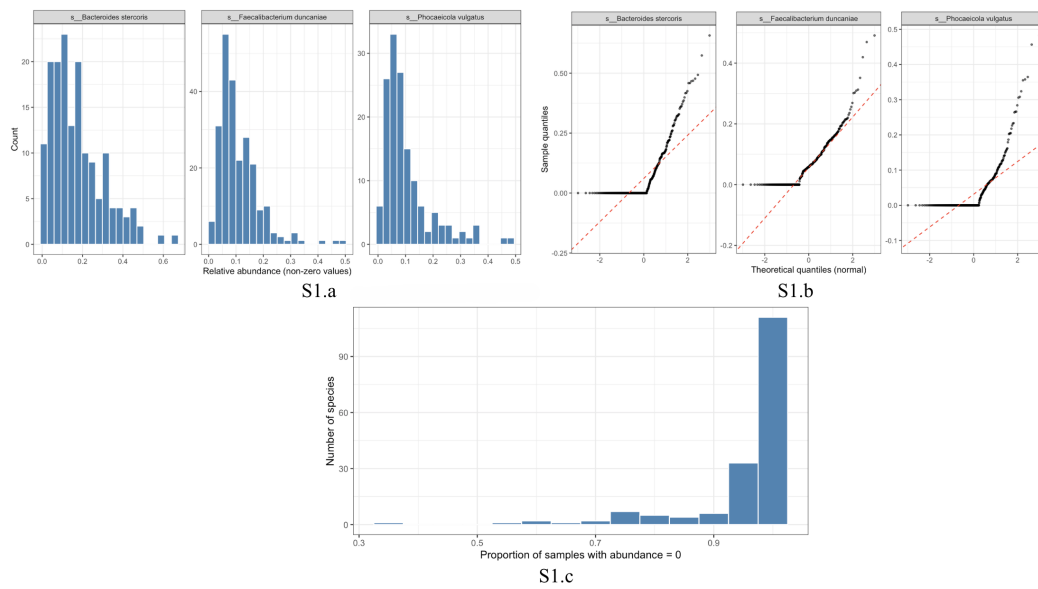


Figure S1: Zero-inflation - Right-skewed distributions

S2:

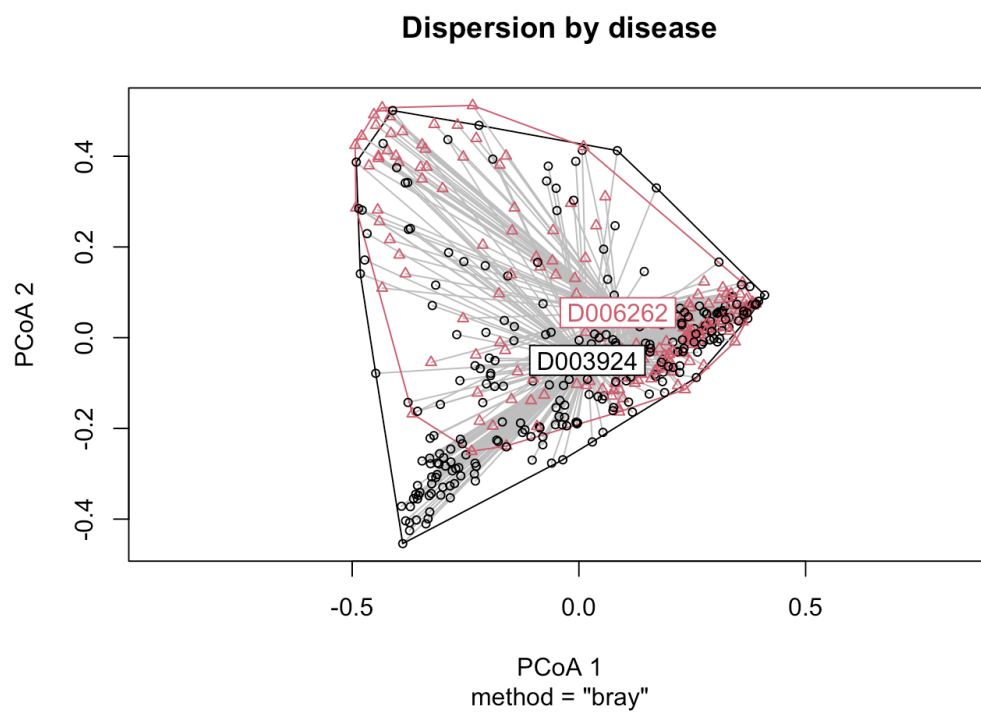


Figure S2: Beta-dispersion plot: Dispersion by disease (full dataset)

S3:

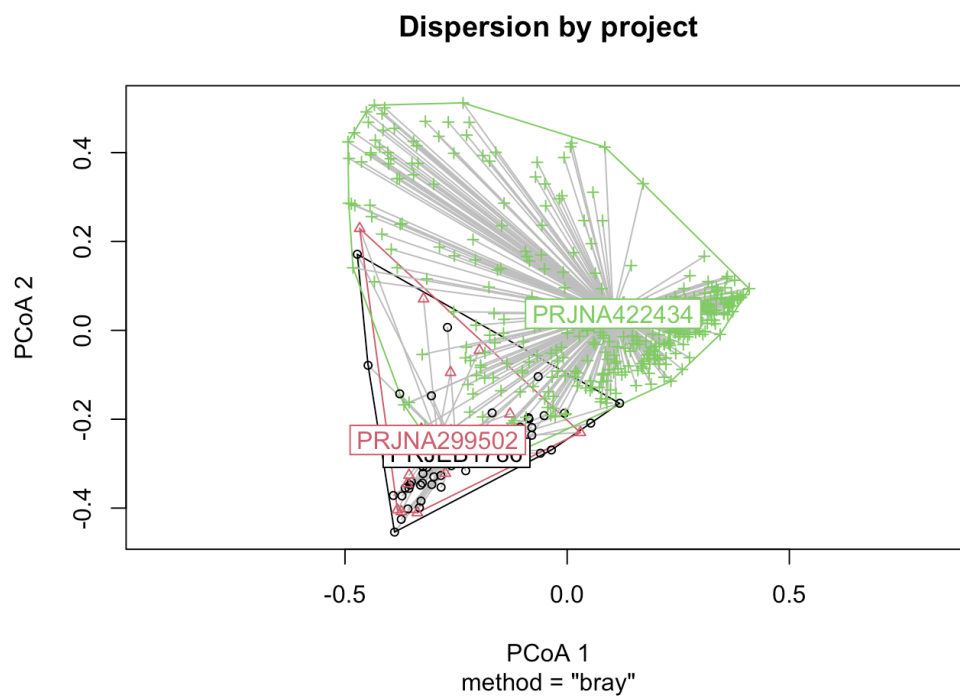


Figure S3: Beta-dispersion plot: Dispersion by project (full dataset)

S4:

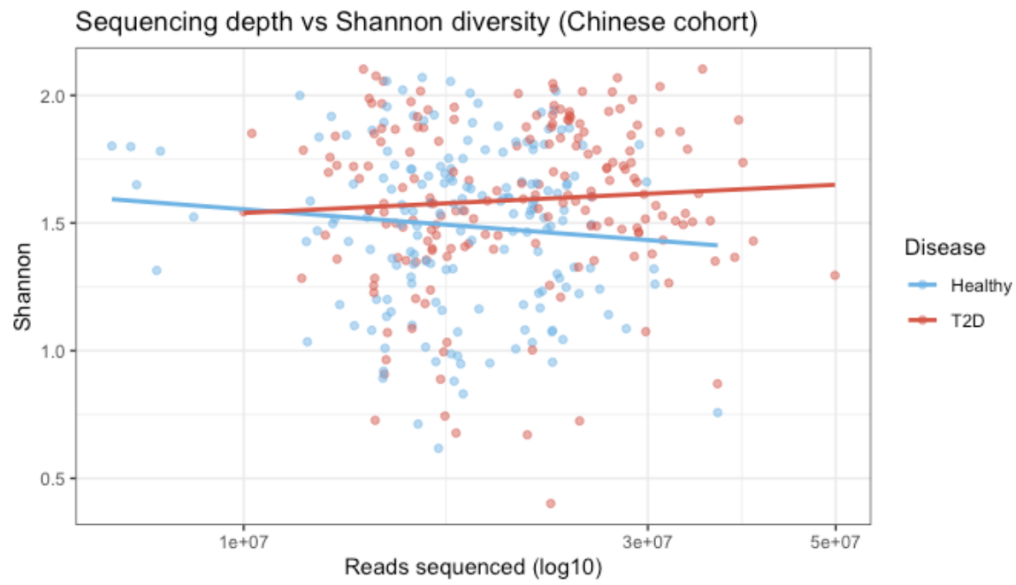


Figure S4: Sequencing depth vs Shannon diversity

S5:

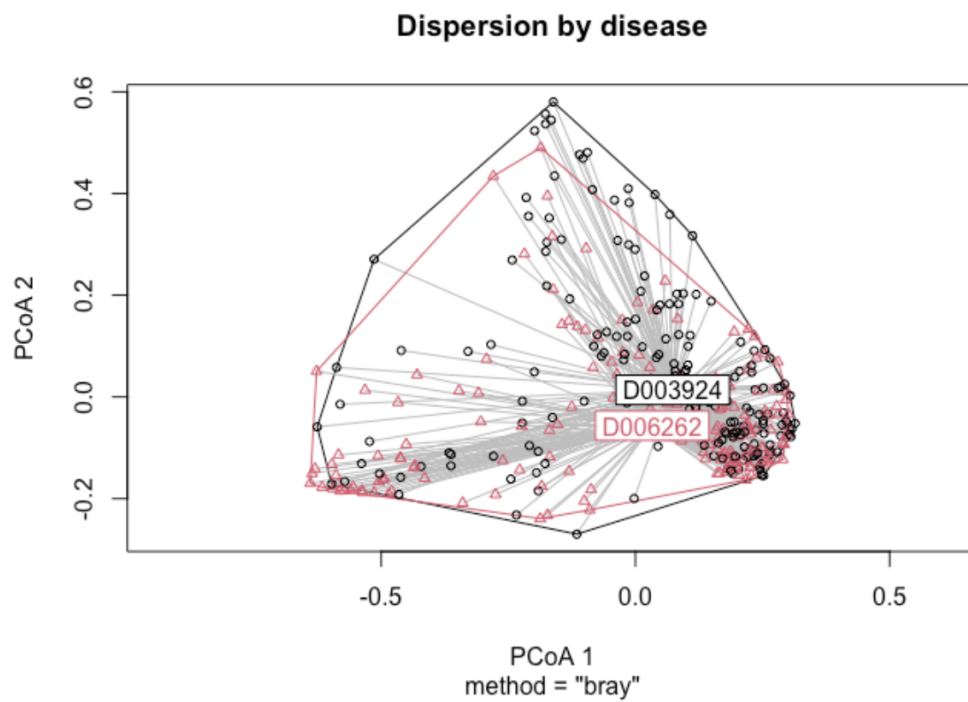


Figure S5: Beta-dispersion plot: Dispersion by disease (Chinese cohort)

S6:

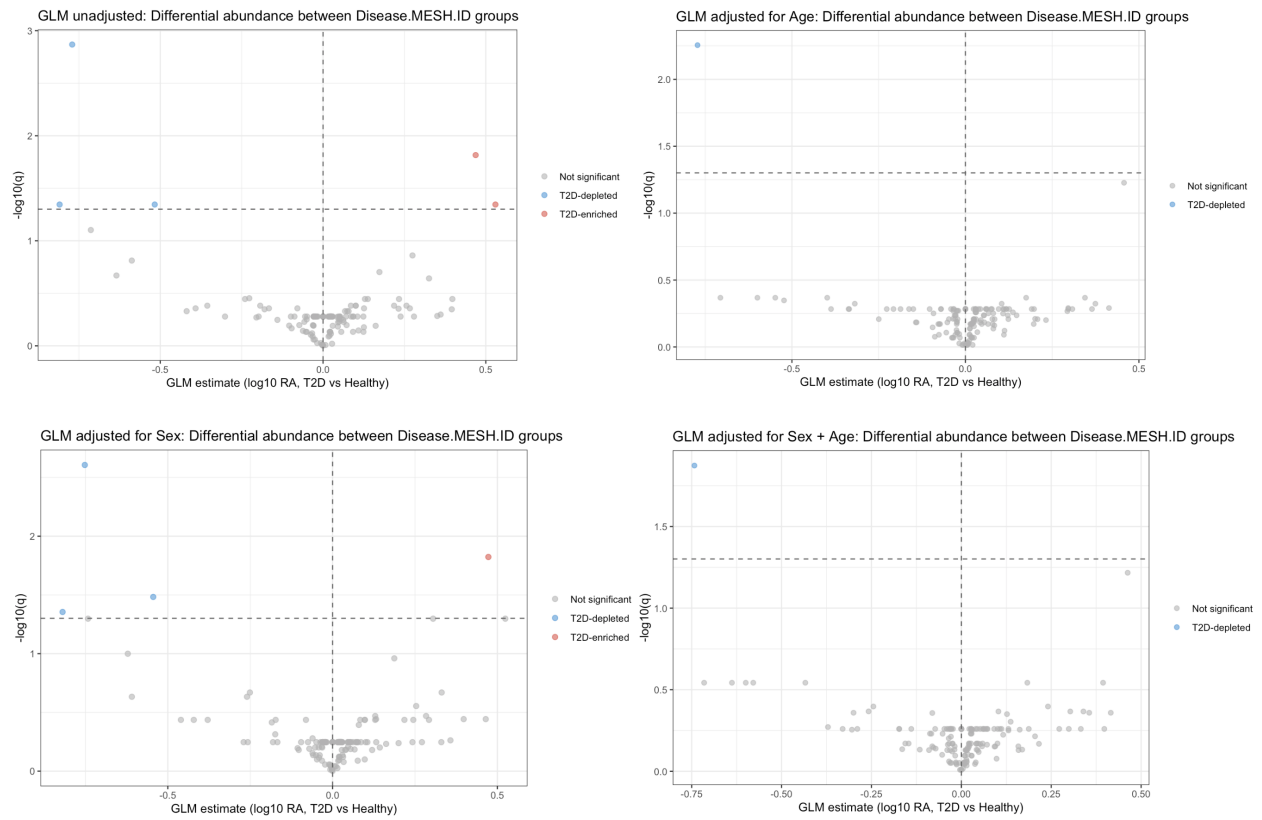


Figure S6: GLM

S7:

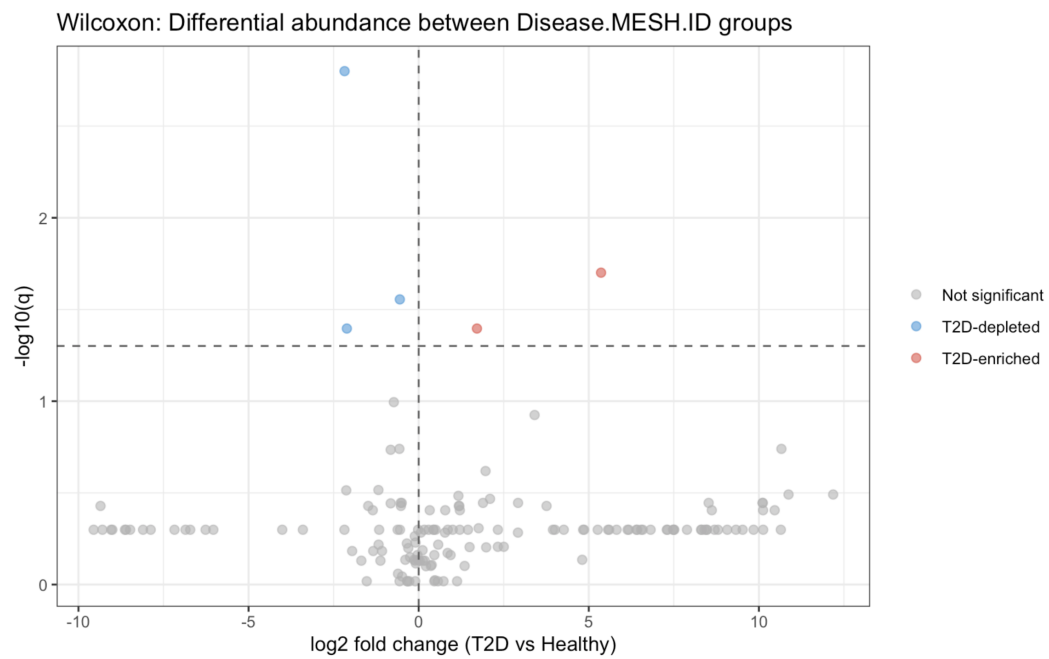


Figure S7: Wilcoxon

S8:

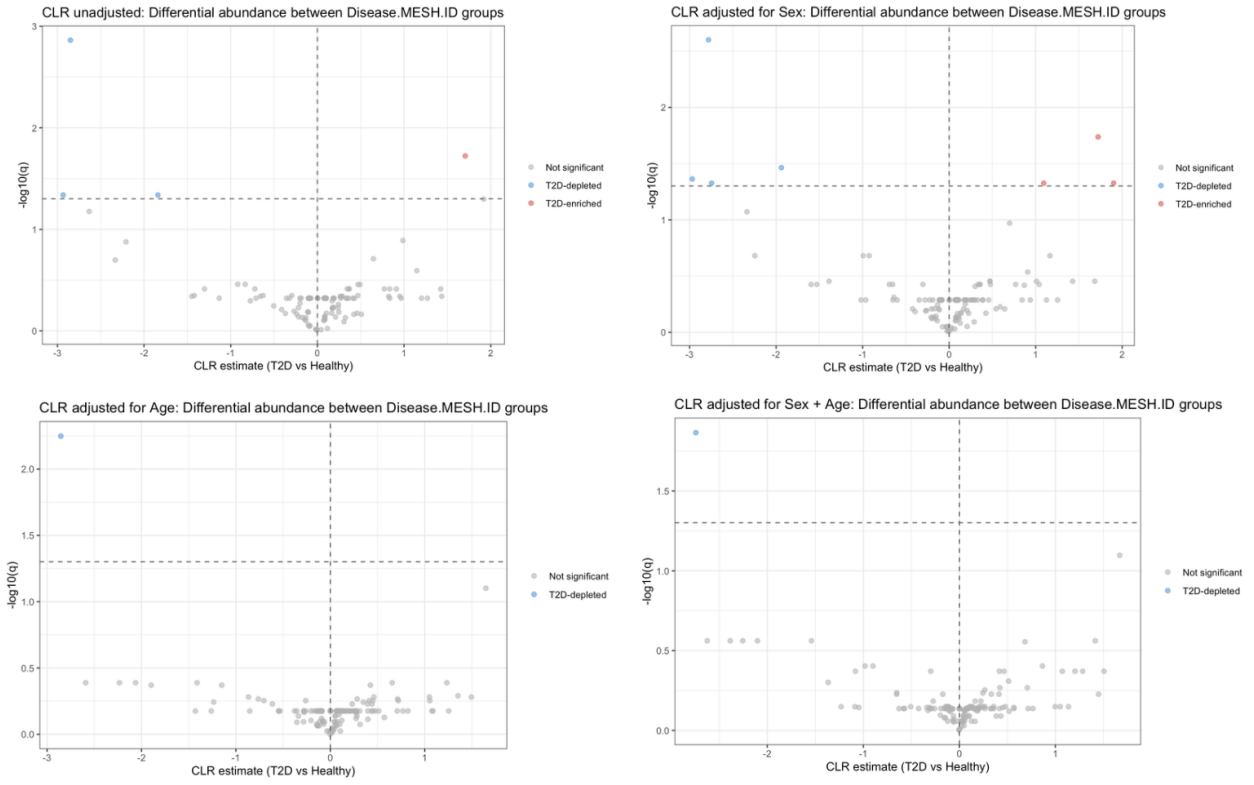


Figure S8: CLR

S9:

Table 1: Unfiltered number of samples in each study group	
Group	Number of samples
T2D (MESH ID: D003924)	260
Healthy controls (MESH ID: D006262)	172

S10:

Project ID	Number of samples
PRJEB1786	54
PRJNA299502	19
PRJNA422434	359
Total	432

Table 2: Number of samples from each project included in the dataset.

S11:

Project ID	T2D	Healthy
PRJEB1786	54	0
PRJNA299502	19	0
PRJNA422434	187	172
Total	260	172

Table 3: Distribution of disease status across projects.